SC1006
Computer Organisation and Architecture

Computer Memory Introduction

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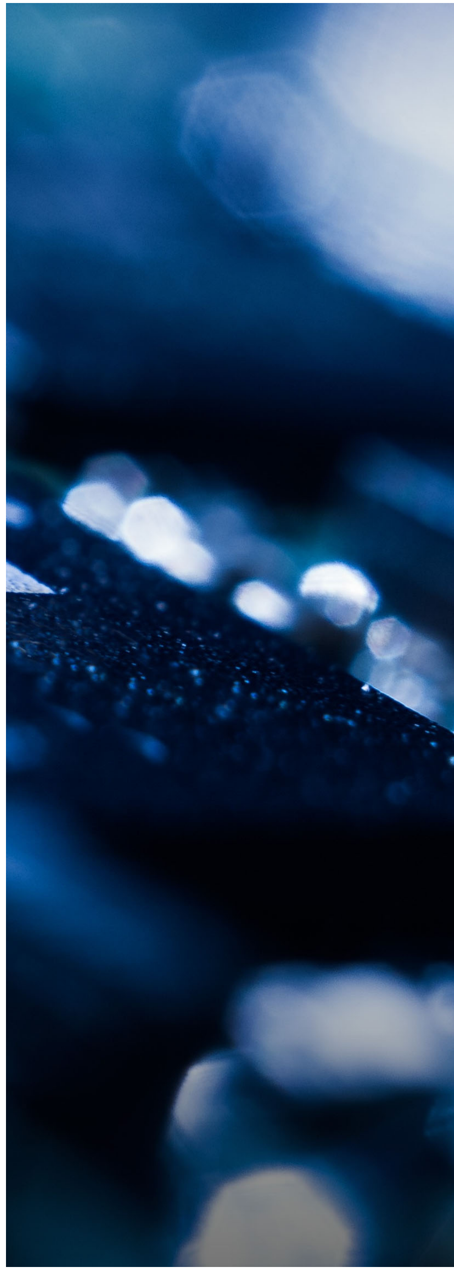
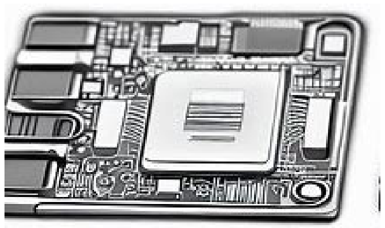


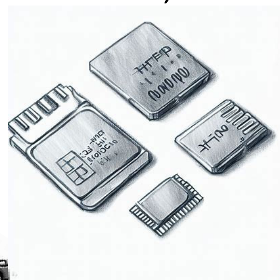
Computer Memory

Semiconductor-based

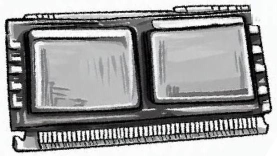
Solid-State Drive

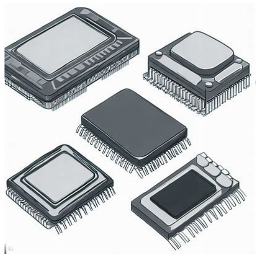


CF, SD, miniSD, microSD, MMC



SODIMM



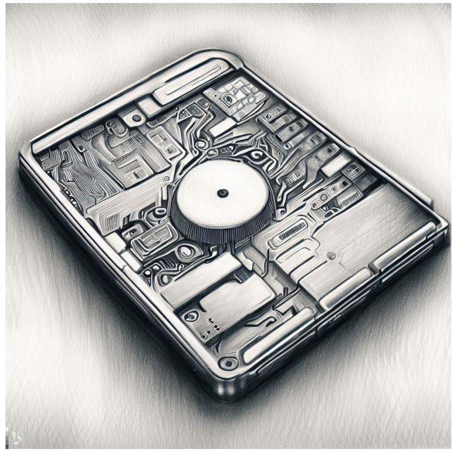


Memory IC Chips

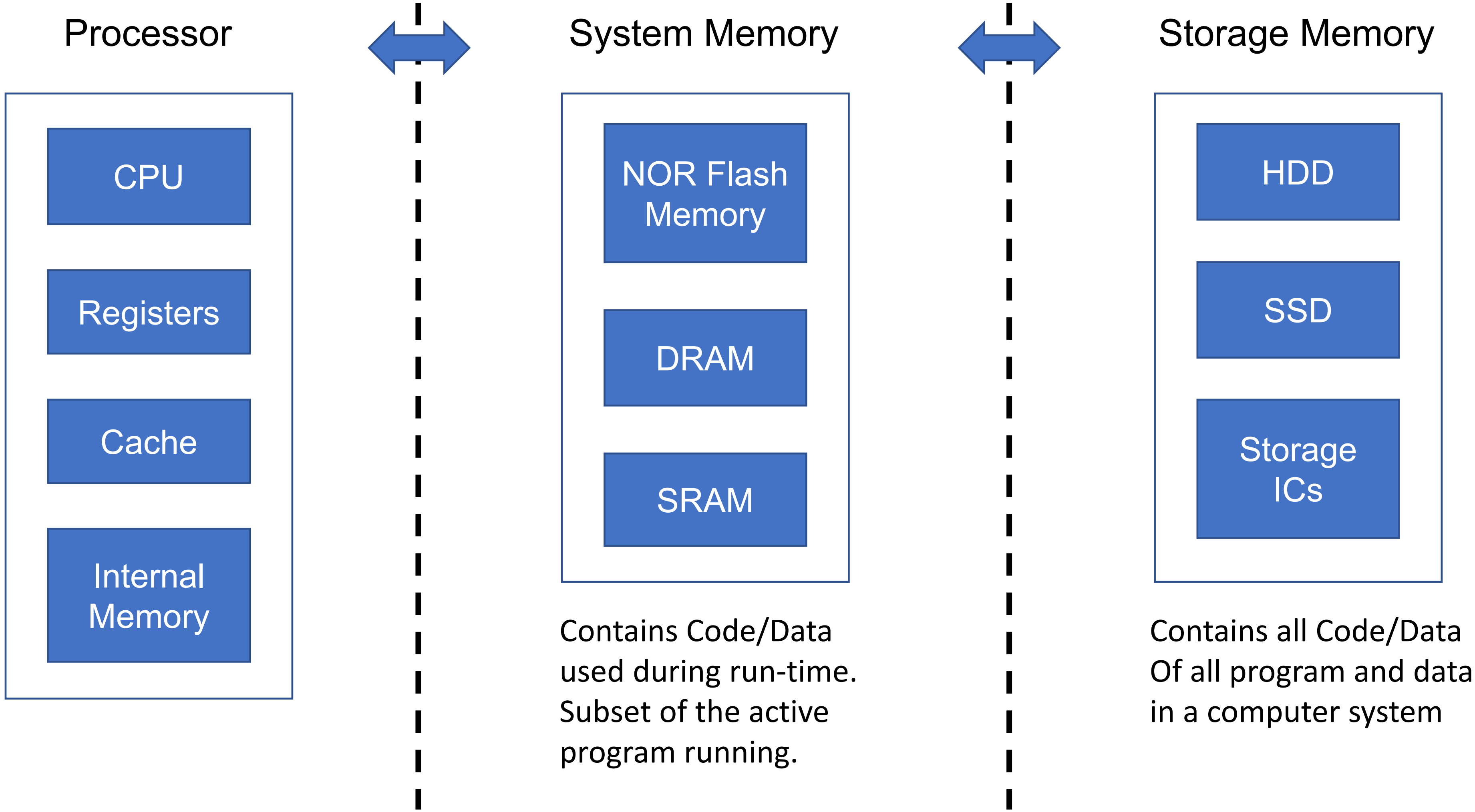
Thumb Drive

Magnetic-based

Magnetic HD

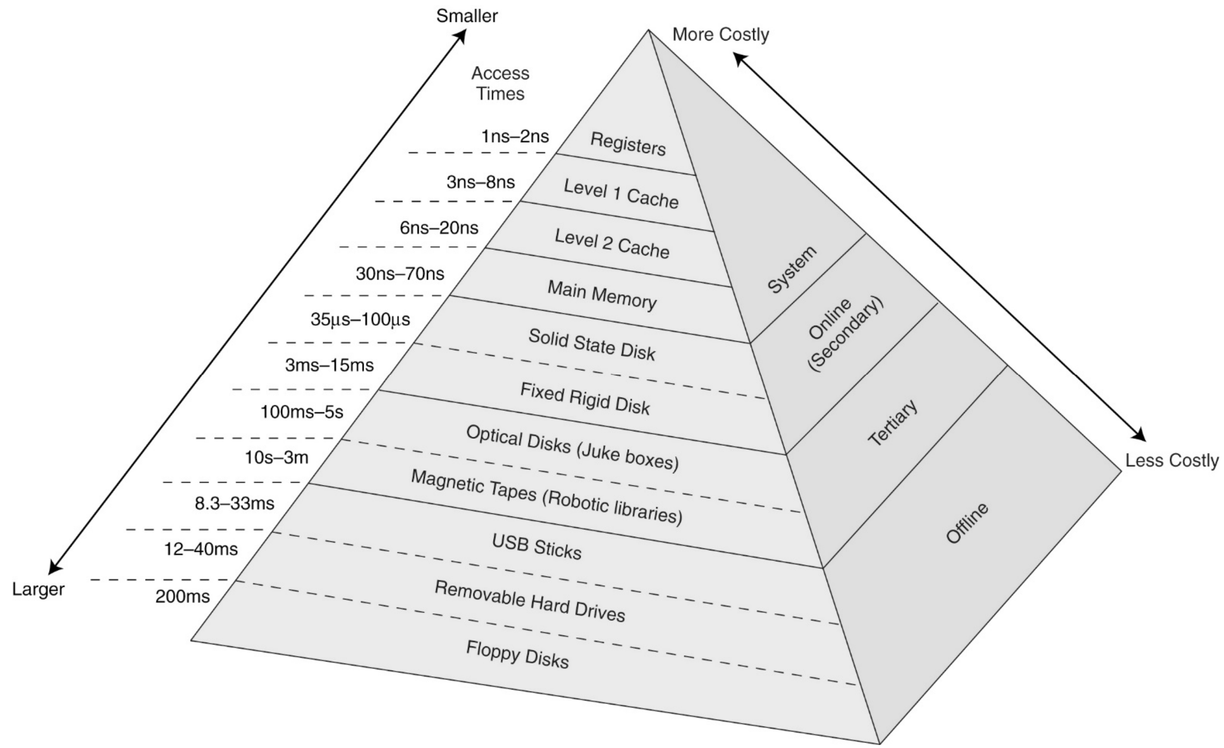


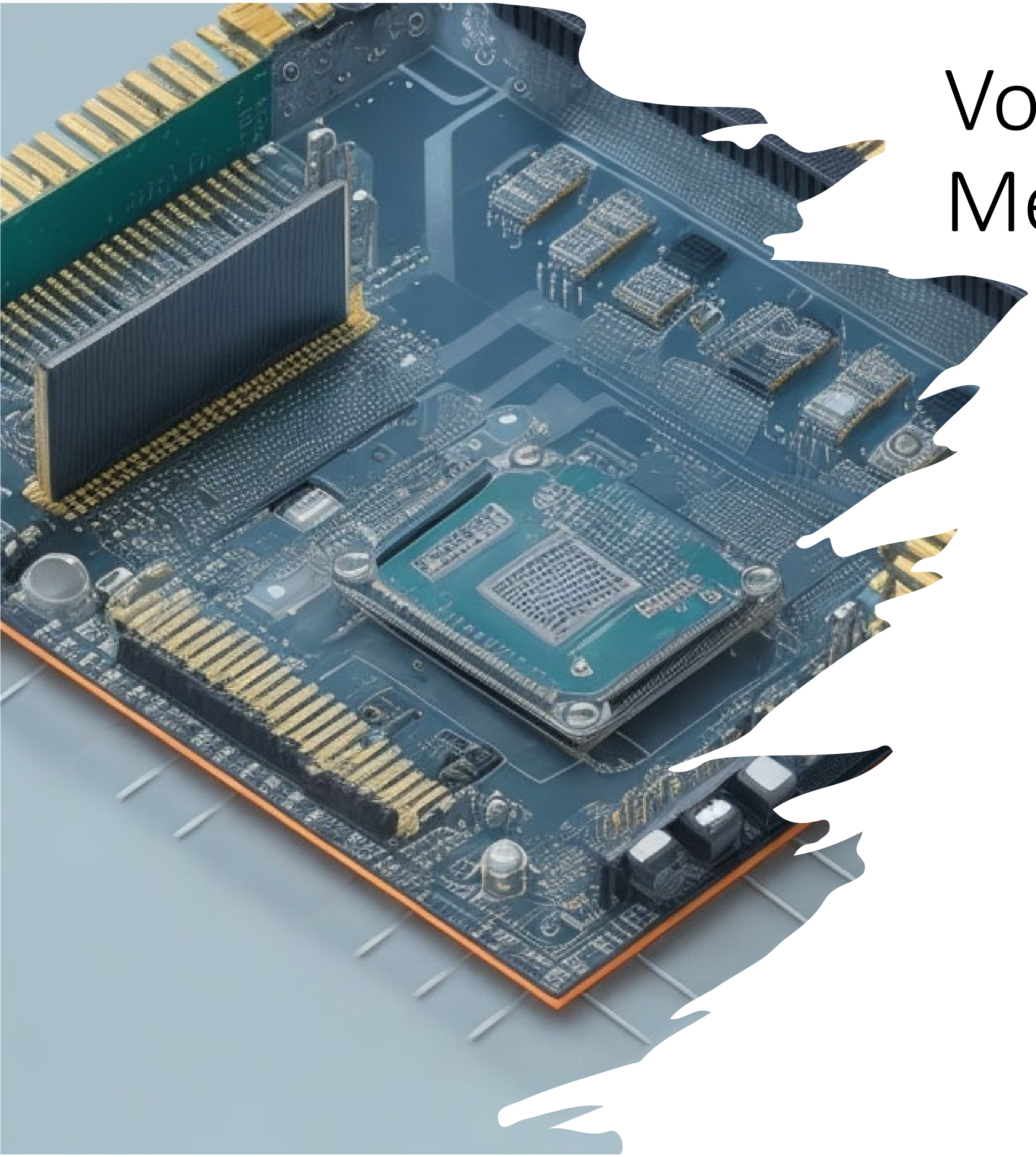
Computer Memory (Functional View)



Memory – Cost vs Function Trade Off

- The memory and storage devices may be organized like a pyramid.
- The pinnacle has the fastest access time but is also more costly.
- Memory Access Time below is only for illustration. These changes with improvement in technology.





Volatile and Non-Volatile Memory

- Volatile
  - Data is lost when electric power is removed.
  - Temporary storage.
  - Typically used as system memory.
  - We will look at Random Access Memories such as Static-RAM (SRAM) and Dynamic-RAM (DRAM).
- Non-volatile
  - Data is retained even if electric power is removed.
  - Permanent storage.
  - Typically used as main storage.
  - We will look at FLASH, magnetic hard-disk specifically.

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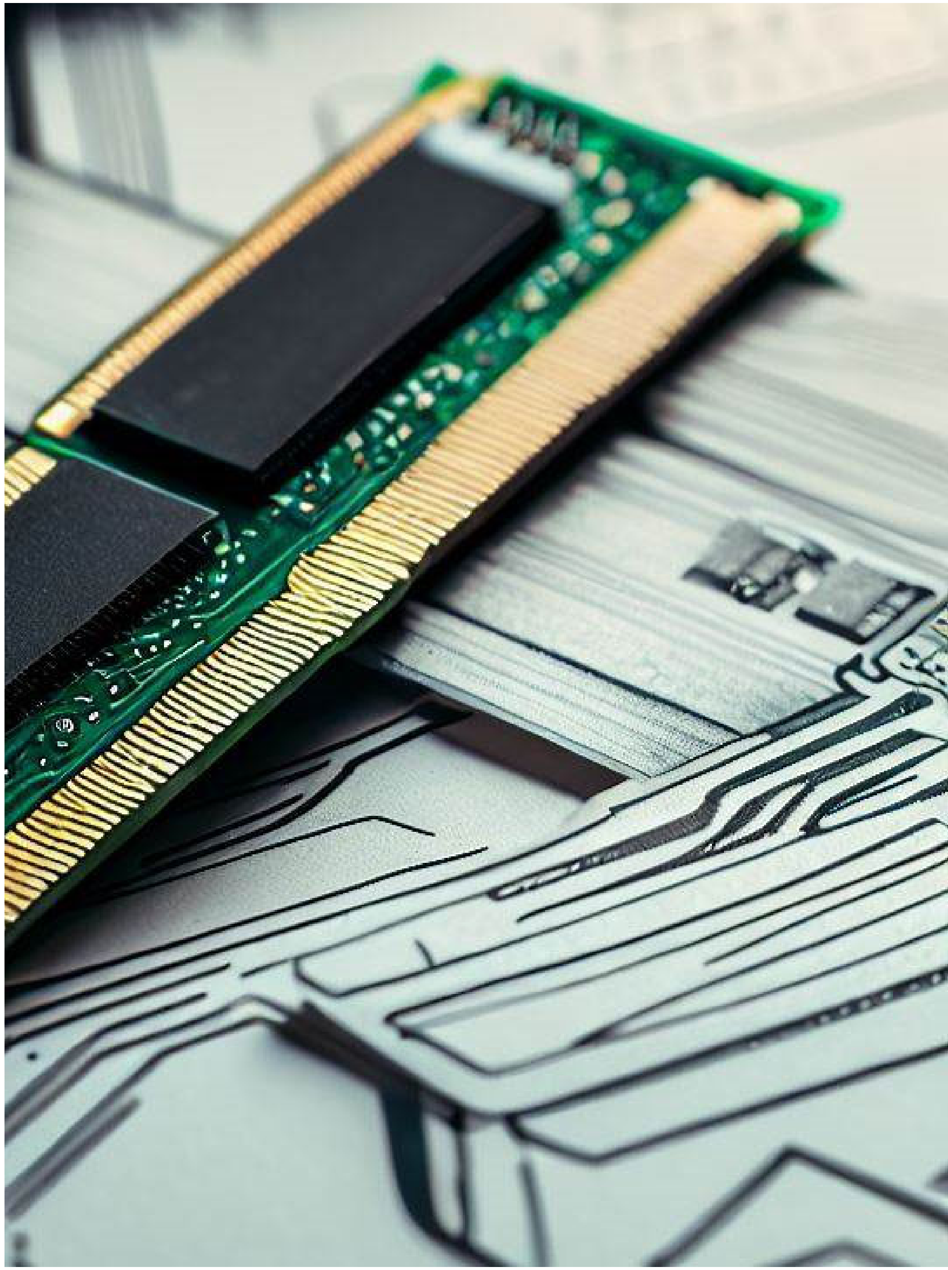

Semiconductor Memories

- Memories based on semiconductor integrated circuits (IC).
- Used as processor internal memories and system memory in a computer system
- Processor internal memories
  - Registers
  - Buffers
  - Cache
  - Internal System and Storage Memory (SRAM, DRAM, Flash based)
- Types of memory
  - SRAM
  - DRAM
  - Flash Memory
  - Solid State Drives (based on Flash Memories)